Lecture Scribe Tutorial 2

Tutorial_2_Solution.pdf
PDF

You are generating a high-quality Lecture Scribe (exam-oriented study guide) strictly using the lecture PPT and tutorial material that I provide.

Context:
This scribe is for closed-notes exam preparation. The content must strictly correspond to the topics and questions covered in the provided lecture PPT and Tutorial (e.g., Tutorial 2 / Lecture 14). Do not introduce any external knowledge, examples, or theory beyond what exists in the provided material.

Core Objective:
Create structured, concept-focused, formula-rich notes that directly correspond to the tutorial questions and lecture topics. The notes must include formulas, derivations, explanations, reasoning steps, and (where relevant) complete worked solutions.

Strict Content Rules:

1. Use only the material present in the uploaded lecture PPT and tutorial document.
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3. You may restate and reorganize content, but do not copy explanatory text verbatim.
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6. If a figure, equation number, or table is referenced (e.g., Equation (3), Fig. 2), explicitly mention it and explain its role in the reasoning.

Structure Requirements:

1. Clear Sectioning
   Organize the scribe topic-wise and question-wise.
   Use headings such as:

* Question 1
* Question 2
* Question 3
  or topic-based headings matching the lecture structure.

2. For Each Question Include:

* Concept Being Tested
* Key Definitions (if present in material)
* Core Formulas Used
* Step-by-step Logical Solution Flow
* Explanation of Why Each Step Works
* Final Result
* Interpretation (if relevant)

3. Formula Emphasis (Very Important)

* Clearly isolate important formulas.
* Show derivation flow where present.
* Highlight dependencies between equations.
* Mention assumptions (independence, distribution type, uniformity, etc.) explicitly.

4. Reasoning Clarity

* Explain transitions between steps.
* Justify substitutions.
* Show why derivatives are taken, why integrals are split, why conditioning is applied, etc.
* Make implicit reasoning explicit.

5. Exam-Focused Enhancements
After each question, add:

* Common mistakes to avoid (based strictly on the material)
* Quick exam tips (e.g., recognize binomial structure after simplification, identify memoryless property, use symmetry, etc.)
* Important pattern recognition hints (only if derivable from the given content)

6. Mathematical Presentation Rules

* Preserve all distributions, parameters, and notation exactly.
* Maintain clean mathematical flow.
* Do not skip intermediate logic if it appears in the original solution.

Phased Output Requirement:

Phase 1:
Generate a fully sectioned Lecture Scribe covering ALL questions/topics present in the provided material.

Phase 2:
Add an Appendix Section containing:

* A compact formula sheet summarizing all key formulas used
* Distribution properties used (Uniform, Exponential, Poisson, Normal, Binomial, etc., if present)
* Important derived expressions
* Conditional probability structures used
* Transformation formulas used
* Any recurring structural results

Tone and Style:

* Clear, structured, precise, and exam-ready.
* Conceptually deep but concise.
* Logical progression over narrative explanation.
* Written for a student revising one day before an exam.

Final Reminder:
Do not introduce any material beyond the uploaded lecture PPT and tutorial. Reorganize, clarify, and enhance - but do not expand beyond the provided content.

Below is the **Lecture Scribe (Exam-Oriented Study Guide)** strictly constructed from **Tutorial-2: Solution** Tutorial_2_Solution.
No external material has been added. Structure follows question-wise organization with full derivation logic preserved.

LECTURE SCRIBE

CSE 400 – Fundamentals of Probability in Computing
Tutorial-2 Solution Based Notes Tutorial_2_Solution

Question 1

Concept Being Tested

Uniform distribution on a continuous interval and geometric probability.

Given

A point is chosen uniformly on a segment of length L.
Let x be the distance from one end.

$$0 < x < L$$

Two segments formed:

- x
- $L - x$

Ratio:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We want:

$$P\left(R < \frac{1}{4}\right)$$

Step-by-Step Logical Solution

Because of symmetry, divide at midpoint $L/2$.

Case 1: $x \leq L/2$

Shorter = x
Longer = $L - x$

$$\frac{x}{L-x} < \frac{1}{4}$$

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > L/2$

Shorter = $L - x$
Longer = x

$$\frac{L-x}{x} < \frac{1}{4}$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$(0, L/5) \cup (4L/5, L)$$

Total length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Uniform distribution over length L:

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Final Result

$$\boxed{P = \frac{2}{5}}$$

Common Mistakes

- Forgetting symmetry.
- Not splitting at midpoint.
- Using incorrect inequality direction.

Exam Tip

Whenever min/max appears with symmetric interval → split at midpoint.

Question 2

Concept Being Tested

Bayes' theorem with continuous densities and equal posterior condition.

Given

$$X|W \sim N(4,4), \quad \sigma_1 = 2$$

$$X|B \sim N(6,9), \quad \sigma_2 = 3$$

$$P(B) = \alpha, \quad P(W) = 1-\alpha$$

Observed $X = 5$

Equal error condition:

$$P(B|X=5) = P(W|X=5) = \frac{1}{2}$$

Apply Bayes' Theorem

$$\frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)} = \frac{1}{2}$$

Normal density:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute densities:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solve:

$$6\alpha = 3\alpha + 2(1-\alpha)e^{-5/72}$$

$$3\alpha = 2(1-\alpha)(0.93)$$

$$4.86\alpha = 1.86$$

$$\boxed{\alpha \approx 0.3827}$$

Common Mistakes

- Forgetting prior probabilities.
- Cancelling exponential incorrectly.

Exam Tip

Equal error ⇒ equal posterior ⇒ numerator = denominator.

Question 3

(a) Uniform Case

Concept

Minimizing expected absolute deviation.

$$X \sim Uniform(0, A)$$

$$f_X(x) = \frac{1}{A}$$

We minimize:

$$E[|X - a|]$$

Break integral:

$$E[|X-a|] = \int_0^a (a-x)\frac{1}{A}dx + \int_a^A (x-a)\frac{1}{A}dx$$

After evaluation:

Derivative:

$$\frac{d}{da}\left(\frac{2a^2 - 2aA + A^2}{2A}\right) = \frac{1}{2A}(4a - 2A)$$

Set to zero:

$$4a = 2A$$

$$\boxed{a = A/2}$$

Station at midpoint.

(b) Exponential Case

$$X \sim Exponential(\lambda)$$

$$f(x) = \lambda e^{-\lambda x}$$

Derivative directly:

$$\frac{d}{da} E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set to zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = 1/2$$

$$a = \frac{\ln 2}{\lambda}$$

Exam Tip

Uniform ⇒ median = midpoint
Exponential ⇒ solve derivative equation directly.

Question 4

Concept

Mixture of exponentials + conditional probability.

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$\boxed{P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}}$$

Exam Tip

Mixture $\neq$ memoryless.

Question 5

Concept

Poisson + conditioning on sum.

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

We want:

$$P(X = k | X + Y = n)$$

$$= \frac{P(X = k)P(Y = n - k)}{P(X + Y = n)}$$

After simplification:

$$\boxed{P(X = k | X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}}$$

Interpretation

Conditional distribution is **Binomial**.

Question 6

Concept

Transformation with mixed discrete-continuous distribution.

$$X \sim Uniform[-2, 2]$$

Piecewise function $g(x)$.

CDF (Piecewise)

$$F_Y(y) = \begin{cases} 0 & y < -2 \\ \frac{y+2}{4} & -2 \leq y < -1 \\ \frac{1}{4} & -1 \leq y < 0 \\ \frac{3}{4} & 0 < y < 1 \\ \frac{y+2}{4} & 1 \leq y \leq 2 \\ 1 & y > 2 \end{cases}$$

Point mass:

$$P(Y=0)=\frac{1}{2}$$

PDF

$$f_Y(y)=\frac{1}{4}\text{ for }(-2,-1),(1,2)$$

Exam Tip

Flat region in CDF ⇒ point mass.

Question 7

Concept

Binomial approximation via logarithmic inequality.

Final price:

$$S_{1000}=S_0u^kd^{1000-k}$$

Condition:

$$u^kd^{1000-k}\geq 1.30$$

Taking logs:

$$k(\ln u-\ln d)+1000\ln d\geq \ln 1.30$$

Solve:

$$k\geq 470$$

$$k\sim Binomial(1000,0.52)$$

Final probability:

$$\boxed{P(k\geq 470)=\sum_{k=470}^{1000}\binom{1000}{k}(0.52)^k(0.48)^{1000-k}}$$

Question 8

$$X\sim Exponential(1/2)$$

PDF:

$$f(x) = \frac{1}{2}e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geq 10 | X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Exam Tip

Exponential ⇒ memoryless ⇒ ratio simplifies.

Question 9

Concept

Transformation $Z = \sqrt{X^2 + Y^2}$

CDF:

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

Region: circle radius z

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiate using Leibniz rule:

$$\boxed{f_Z(z) = z \int_{-z}^{z} \frac{f_X(x)\left[f_Y(\sqrt{z^2-x^2}) + f_Y(-\sqrt{z^2-x^2})\right]}{\sqrt{z^2-x^2}} dx}$$

Question 10

Concept

Joint distribution of maxima.

$$M_n = \max(X_1, ..., X_n)$$

$$P(M_n \leq x) = F(x)^n$$

Joint CDF:

$$P(M_n \le x, M_{n+1} \le y) = \begin{cases} F(x)^n F(y) & x \le y \\ F(y)^{n+1} & x > y \end{cases}$$

PHASE 2 — APPENDIX

Core Formula Sheet

Uniform PDF

$$f(x) = \frac{1}{A}$$

Exponential PDF

$$f(x) = \lambda e^{-\lambda x}$$

Memoryless Property

$$P(X > t + s | X > t) = P(X > s)$$

Normal Density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Poisson PMF

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Binomial PMF

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Important Structural Results

- Conditional Poisson ⇒ Binomial
- Mixture exponential ≠ memoryless
- Flat CDF ⇒ discrete mass
- Max of i.i.d. ⇒ power of CDF
- Log transform simplifies multiplicative inequalities

This completes the full structured lecture scribe strictly based on Tutorial-2 solution Tutorial_2_Solution.

If you'd like, I can now:

- Convert this into a compact 5-page printable revision sheet
- Or generate high-probability exam question patterns from this tutorial only

Sources

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Below is the **Fully Structured Lecture Scribe (Exam-Oriented, Closed-Notes Ready)** constructed **strictly from Tutorial-2: Solution** Tutorial_2_Solution.
No external material has been introduced. All derivations, formulas, and reasoning steps follow the provided document.

LECTURE SCRIBE

CSE 400 – Fundamentals of Probability in Computing
Tutorial-2 Structured Study Notes Tutorial_2_Solution

PHASE 1 — FULL QUESTION-WISE SCRIBE

Question 1

Concept Being Tested

Geometric probability under **Uniform distribution** on an interval.
Handling **min/max expressions** via case analysis.

Problem Structure

A point is chosen uniformly on a line segment of length L.

Let:

$$0 < x < L$$

Segments formed:

- x
- $L - x$

Ratio:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We need:

$$P\left(R < \frac{1}{4}\right)$$

Core Reasoning

Because the minimum switches at the midpoint, divide at:

$$x = \frac{L}{2}$$

Case 1: $x \leq \frac{L}{2}$

Shorter = x
Longer = $L - x$

$$\frac{x}{L-x} < \frac{1}{4}$$

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

Shorter = $L - x$
Longer = x

$$\frac{L-x}{x} < \frac{1}{4}$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$(0, L/5) \cup (4L/5, L)$$

Total length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since uniform:

$$P = \frac{\text{valid length}}{\text{total length}} = \frac{2L/5}{L} = \boxed{\frac{2}{5}}$$

Common Mistakes

- Not splitting at midpoint.
- Forgetting uniform ⇒ probability = length ratio.

Pattern Recognition

Min/max on symmetric interval ⇒ split at midpoint.

Question 2

Concept Being Tested

Bayes' Theorem with continuous densities.
Equal posterior condition for equal error.

Given

$$X|W \sim N(4,4), \quad \sigma_1 = 2$$

$$X|B \sim N(6,9), \quad \sigma_2 = 3$$

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observation: $X = 5$

Equal error condition:

$$P(B|X=5) = P(W|X=5) = \frac{1}{2}$$

Bayes' Formula

$$P(B|X=5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)}$$

Set equal to $1/2$.

Normal Density Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitution leads to:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solve

$$6\alpha = 3\alpha + 2(1-\alpha)e^{-5/72}$$

$$3\alpha = 2(1-\alpha)(0.93)$$

$$4.86\alpha = 1.86$$

$$\boxed{\alpha \approx 0.3827}$$

Common Mistakes

- Forgetting priors.
- Cancelling exponential terms incorrectly.

Exam Tip

Equal posterior ⇒ numerator = denominator.

Question 3

(a) Uniform Case

Concept

Minimizing:

$$E[|X-a|]$$

$$X \sim Uniform(0, A)$$

$$f_X(x) = \frac{1}{A}$$

Split Integral

$$E[|X-a|] = \int_0^a (a-x)\frac{1}{A}dx + \int_a^A (x-a)\frac{1}{A}dx$$

After evaluation:

$$\frac{2a^2 - 2aA + A^2}{2A}$$

Differentiate:

$$\frac{d}{da} = \frac{1}{2A}(4a - 2A)$$

Set to zero:

$$4a = 2A$$

$$\boxed{a = \frac{A}{2}}$$

(b) Exponential Case

$$X \sim Exponential(\lambda)$$

$$f(x) = \lambda e^{-\lambda x}$$

$$E[|X-a|] = \int_0^a (a-x)\lambda e^{-\lambda x}dx + \int_a^\infty (x-a)\lambda e^{-\lambda x}dx$$

Derivative directly:

$$\frac{d}{da}E[|X-a|] = 1 - 2e^{-\lambda a}$$

Set to zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = 1/2$$

$$\boxed{a = \frac{\ln 2}{\lambda}}$$

Key Insight

Uniform ⇒ midpoint.
Exponential ⇒ solve derivative equation.

Question 4

Concept

Mixture of exponential distributions.
Law of total probability.

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$\boxed{P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}}$$

Important

Mixture distribution ≠ memoryless.

Question 5

Concept

Conditioning Poisson on total count.

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

$$P(X = k | X + Y = n) = \frac{P(X = k) P(Y = n - k)}{P(X + Y = n)}$$

After simplification:

$$\boxed{P(X = k | X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}}$$

Structural Result

Conditional distribution becomes **Binomial**.

Question 6

Concept

Transformation of Uniform RV.
Mixed continuous + discrete outcome.

$$X \sim Uniform[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

CDF (Piecewise)

$$F_Y(y) = \begin{cases} 0 & y < -2 \\ \frac{y+2}{4} & -2 \leq y < -1 \\ \frac{1}{4} & -1 \leq y < 0 \\ \frac{3}{4} & 0 < y < 1 \\ \frac{y+2}{4} & 1 \leq y \leq 2 \\ 1 & y > 2 \end{cases}$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

PDF

$$f_Y(y) = \frac{1}{4} \text{ for } (-2, -1), (1, 2)$$

Key Pattern

Flat region in CDF ⇒ discrete mass.

Question 7

Concept

Binomial modeling after log transformation.

Final price:

$$S_{1000} = S_0 u^k d^{1000-k}$$

Condition:

$$u^k d^{1000-k} \geq 1.30$$

Take logs:

$$k(\ln u - \ln d) + 1000 \ln d \geq \ln(1.30)$$

Solve:

$$k \geq 470$$

$$k \sim Binomial(1000, 0.52)$$

$$\boxed{P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}}$$

Question 8

$$X \sim Exponential(1/2)$$

$$f(x) = \frac{1}{2} e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geq 10 | X > 9) = \frac{e^{-5}}{e^{-9/2}} = \boxed{e^{-1/2}}$$

Question 9

Concept

Transformation $Z = \sqrt{X^2 + Y^2}$

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiate:

$$\frac{d}{dz}\sqrt{z^2-x^2}=\frac{z}{\sqrt{z^2-x^2}}$$

Final PDF:

$$\boxed{f_Z(z)=z\int_{-z}^{z}\frac{f_X(x)\left[f_Y(\sqrt{z^2-x^2})+f_Y(-\sqrt{z^2-x^2})\right]}{\sqrt{z^2-x^2}}dx}$$

Question 10

Concept

Distribution of maxima.

$$M_n=\max(X_1,\ldots,X_n)$$

$$P(M_n\le x)=F(x)^n$$

Joint CDF:

$$P(M_n\le x, M_{n+1}\le y)=\begin{cases}F(x)^nF(y), & x\le y\\ F(y)^{n+1}, & x>y\end{cases}$$

PHASE 2 — APPENDIX

Compact Formula Sheet

Uniform

$$f(x)=\frac{1}{A}$$

Exponential

$$f(x)=\lambda e^{-\lambda x}$$

Normal

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Poisson

$$P(X=k)=e^{-\lambda}\frac{\lambda^k}{k!}$$

Binomial

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$$

Structural Results Used

- Length ratio for Uniform.
- Bayes posterior equality.
- Minimization via derivative.
- Law of total probability.
- Conditional Poisson ⇒ Binomial.
- Mixed distribution from transformation.
- Log transform converts product inequality to linear inequality.
- Max of i.i.d. ⇒ power of CDF.

This completes the fully structured, exam-focused Lecture Scribe strictly based on Tutorial-2 Tutorial_2_Solution.

If you'd like, I can now compress this into a 3-page ultra-compact revision sheet for last-night preparation.

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* Make implicit reasoning explicit.

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   After each question, add:

* Common mistakes to avoid (based strictly on the material)
* Quick exam tips (e.g., recognize binomial structure after simplification, identify memoryless property, use symmetry, etc.)
* Important pattern recognition hints (only if derivable from the given content)

6. Mathematical Presentation Rules

* Preserve all distributions, parameters, and notation exactly.
* Maintain clean mathematical flow.
* Do not skip intermediate logic if it appears in the original solution.

Phased Output Requirement:

Phase 1:
Generate a fully sectioned Lecture Scribe covering ALL questions/topics present in the provided material.

Phase 2:
Add an Appendix Section containing:

* A compact formula sheet summarizing all key formulas used
* Distribution properties used (Uniform, Exponential, Poisson, Normal, Binomial, etc., if present)
* Important derived expressions
* Conditional probability structures used
* Transformation formulas used
* Any recurring structural results

Tone and Style:

* Clear, structured, precise, and exam-ready.
* Conceptually deep but concise.
* Logical progression over narrative explanation.
* Written for a student revising one day before an exam.

Final Reminder:
Do not introduce any material beyond the uploaded lecture PPT and tutorial. Reorganize, clarify, and enhance - but do not expand beyond the provided content.

Below is the **Complete Exam-Oriented Lecture Scribe** constructed **strictly and exclusively** from *Tutorial-2: Solution* Tutorial_2_Solution.
No external theory or examples have been introduced. All formulas, distributions, derivations, and logical transitions follow the provided material.

CSE 400 – Fundamentals of Probability in Computing

Tutorial-2 Lecture Scribe (Closed-Notes Exam Version) Tutorial_2_Solution

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PHASE 1 — FULL STRUCTURED SCRIBE

Question 1

Concept Being Tested

Uniform random selection on an interval and probability via geometric length ratio.
Handling expressions involving $\min$ and $\max$ by case analysis.

Problem Interpretation

A point is chosen uniformly on a line segment of length L.

Let:

$$0 < x < L$$

Segments formed:

- x
- $L - x$

Ratio:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We need:

$$P\left(R < \frac{1}{4}\right)$$

Since the point is uniformly chosen over length L, probability equals:

$$\frac{\text{length of valid region}}{L}$$

Step 1: Split at Midpoint

The minimum switches at:

$$x = \frac{L}{2}$$

Case 1: $x \leq \frac{L}{2}$

Shorter = x
Longer = $L - x$

$$\frac{x}{L-x} < \frac{1}{4}$$

Multiply both sides by $4(L-x)$:

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

Shorter = $L - x$
Longer = x

$$\frac{L-x}{x} < \frac{1}{4}$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Step 2: Combine Regions

Valid region:

$$(0, L/5) \cup (4L/5, L)$$

Total length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Final Probability

$$P = \frac{2L/5}{L} = \boxed{\frac{2}{5}}$$

Common Mistakes

- Not splitting at midpoint.
- Forgetting uniform ⇒ probability equals length ratio.

Exam Recognition Pattern

Min/max over symmetric interval ⇒ split at midpoint.

Question 2

Concept Being Tested

Bayes' Theorem with continuous densities.
Equal posterior condition for equal classification error.

Given

$$X|W \sim N(4,4), \quad \sigma_1 = 2$$

$$X|B \sim N(6,9), \quad \sigma_2 = 3$$

$$P(B) = \alpha, \quad P(W) = 1-\alpha$$

Observation: $X = 5$

Equal error condition:

$$P(B|X=5) = P(W|X=5) = \frac{1}{2}$$

Bayes' Formula

$$P(B|X=5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)}$$

Set equal to $1/2$.

Normal Density Formula Used

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute densities:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solve for α

$$6\alpha = 3\alpha + 2(1-\alpha)e^{-5/72}$$

$$3\alpha = 2(1-\alpha)(0.93)$$

$$4.86\alpha = 1.86$$

$$\boxed{\alpha \approx 0.3827}$$

Common Mistakes

- Ignoring prior probabilities.
- Cancelling exponential terms incorrectly.

Recognition Pattern

Equal posterior ⇒ numerator equals denominator.

Question 3

(a) Uniform Distribution Case

Concept

Minimizing expected absolute distance.

$$X \sim Uniform(0, A)$$

$$f_X(x) = \frac{1}{A}$$

Minimize:

$$E[|X-a|]$$

Split Absolute Value

$$E[|X-a|] = \int_0^a (a-x)\frac{1}{A}dx + \int_a^A (x-a)\frac{1}{A}dx$$

After evaluation:

$$\frac{2a^2 - 2aA + A^2}{2A}$$

Differentiate:

$$\frac{d}{da} = \frac{1}{2A}(4a - 2A)$$

Set derivative = 0:

$$4a = 2A$$

$$\boxed{a = \frac{A}{2}}$$

Station at midpoint.

(b) Exponential Distribution Case

$$X \sim Exponential(\lambda)$$

$$f(x) = \lambda e^{-\lambda x}$$

$$E[|X - a|] = \int_0^a (a - x)\lambda e^{-\lambda x} dx + \int_a^\infty (x - a)\lambda e^{-\lambda x} dx$$

Derivative directly:

$$\frac{d}{da} E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set equal to zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = \frac{1}{2}$$

$$\boxed{a = \frac{\ln 2}{\lambda}}$$

Key Insight

Uniform ⇒ midpoint.
Exponential ⇒ solve derivative equation.

Question 4

Concept

Mixture of exponentials using Law of Total Probability.

Let battery type i have rate λ_i, probability p_i.

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$\boxed{P(X > t+s | X > t) = \frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}}$$

Important Structural Observation

Mixture distribution does **not** reduce to exponential form.

Question 5

Concept

Conditional Poisson distribution.

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

We want:

$$P(X = k | X + Y = n)$$

Using independence:

$$= \frac{P(X = k)P(Y = n - k)}{P(X + Y = n)}$$

After simplifying:

$$\boxed{P(X = k | X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}}$$

Recognition Pattern

Conditional Poisson ⇒ Binomial structure.

Question 6

Concept

Transformation of Uniform variable.
Mixed discrete-continuous distribution.

$$X \sim Uniform[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

CDF

$$F_Y(y) = \begin{cases} 0 & y < -2 \\ \frac{y+2}{4} & -2 \le y < -1 \\ \frac{1}{4} & -1 \le y < 0 \\ \frac{3}{4} & 0 < y < 1 \\ \frac{y+2}{4} & 1 \le y \le 2 \\ 1 & y > 2 \end{cases}$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

PDF

$$f_Y(y) = \frac{1}{4} \quad \text{for } (-2, -1), (1, 2)$$

Recognition Pattern

Flat CDF segment ⇒ point mass.

Question 7

Concept

Log transformation + Binomial modeling.

$$S_{1000} = S_0 u^k d^{1000-k}$$

Condition:

$$u^k d^{1000-k} \ge 1.30$$

Take natural log:

$$k(\ln u - \ln d) + 1000 \ln d \ge \ln(1.30)$$

Solve:

$$k \ge 470$$

$$k \sim Binomial(1000, 0.52)$$

$$\boxed{P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}}$$

Question 8

$$X \sim Exponential(1/2)$$

$$f(x) = \frac{1}{2} e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geq 10 | X > 9) = \frac{e^{-5}}{e^{-9/2}} = \boxed{e^{-1/2}}$$

Question 9

Concept

Transformation:

$$Z = \sqrt{X^2 + Y^2}$$

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

Region: circle radius z

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiate using:

$$\frac{d}{dz} \sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final density:

$$\boxed{f_Z(z) = z \int_{-z}^{z} \frac{f_X(x) \left[f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2}) \right]}{\sqrt{z^2 - x^2}} dx}$$

Question 10

Concept

Distribution of maxima.

$$M_n = \max(X_1, \ldots, X_n)$$

$$P(M_n \leq x) = F(x)^n$$

Joint CDF:

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y \\ F(y)^{n+1}, & x > y \end{cases}$$

PHASE 2 — APPENDIX

Compact Formula Sheet

Uniform

$$f(x) = \frac{1}{A}$$

Exponential

$$f(x) = \lambda e^{-\lambda x}$$

Normal

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Poisson

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Binomial

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Derived Structural Results Used

- Uniform probability via length ratio.
- Bayes posterior equality condition.
- Minimization via derivative.

- Law of Total Probability for mixture.
- Conditional Poisson ⇒ Binomial.
- Mixed distribution from transformation.
- Log transformation for multiplicative inequalities.
- Max of i.i.d. ⇒ power of CDF.

This completes the complete, structured, closed-notes lecture scribe strictly based on Tutorial-2 .

Tutorial_2_Solution

If you'd like, I can now generate a hyper-condensed "Last 2-Hour Revision Sheet" from this exact content only.

Sources

Tutorial_2_Solution.pdf
PDF

You are generating a high-quality Lecture Scribe (exam-oriented study guide) strictly using the lecture PPT and tutorial material that I provide.

Context:
This scribe is for closed-notes exam preparation. The content must strictly correspond to the topics and questions covered in the provided lecture PPT and Tutorial (e.g., Tutorial 2 / Lecture 14). Do not introduce any external knowledge, examples, or theory beyond what exists in the provided material.

Core Objective:
Create structured, concept-focused, formula-rich notes that directly correspond to the tutorial questions and lecture topics. The notes must include formulas, derivations, explanations, reasoning steps, and (where relevant) complete worked solutions.

Strict Content Rules:

1. Use only the material present in the uploaded lecture PPT and tutorial document.
2. Do not add external examples, extra theory, or content from memory.
3. You may restate and reorganize content, but do not copy explanatory text verbatim.
4. Problem statements, definitions, and equation labels may remain unchanged if necessary.
5. All formulas and mathematical expressions must be included exactly as used in the material.
6. If a figure, equation number, or table is referenced (e.g., Equation (3), Fig. 2), explicitly mention it and explain its role in the reasoning.

Structure Requirements:

1. Clear Sectioning
   Organize the scribe topic-wise and question-wise.
   Use headings such as:

* Question 1

* Question 2
* Question 3
  or topic-based headings matching the lecture structure.

2. For Each Question Include:

* Concept Being Tested
* Key Definitions (if present in material)
* Core Formulas Used
* Step-by-step Logical Solution Flow
* Explanation of Why Each Step Works
* Final Result
* Interpretation (if relevant)

3. Formula Emphasis (Very Important)

* Clearly isolate important formulas.
* Show derivation flow where present.
* Highlight dependencies between equations.
* Mention assumptions (independence, distribution type, uniformity, etc.) explicitly.

4. Reasoning Clarity

* Explain transitions between steps.
* Justify substitutions.
* Show why derivatives are taken, why integrals are split, why conditioning is applied, etc.
* Make implicit reasoning explicit.

5. Exam-Focused Enhancements
   After each question, add:

* Common mistakes to avoid (based strictly on the material)
* Quick exam tips (e.g., recognize binomial structure after simplification, identify memoryless property, use symmetry, etc.)
* Important pattern recognition hints (only if derivable from the given content)

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No external theory, examples, or additional results have been introduced.

CSE 400 – Fundamentals of Probability in Computing

Tutorial-2 Exam-Oriented Lecture Scribe Tutorial_2_Solution

PHASE 1 — FULLY SECTIONED SCRIBE

Question 1

Concept Being Tested

Uniform selection on an interval and probability via geometric length ratio.
Handling expressions involving $\min$ and $\max$ through case analysis.

Interpretation

A point is chosen uniformly on a segment of length L.

Let:

$$0 < x < L$$

Segments formed:

- x
- $L - x$

Ratio:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We compute:

$$P\left(R < \frac{1}{4}\right)$$

Since selection is uniform:

$$P = \frac{\text{length of valid region}}{L}$$

Case Analysis (Split at Midpoint)

Switch occurs at:

$$x = \frac{L}{2}$$

Case 1: $x \leq \frac{L}{2}$

Shorter = x
Longer = $L - x$

$$\frac{x}{L-x} < \frac{1}{4}$$

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

Shorter = $L - x$
Longer = x

$$\frac{L-x}{x} < \frac{1}{4}$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$(0, L/5) \cup (4L/5, L)$$

Total length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Final Result

$$\boxed{P = \frac{2}{5}}$$

Common Mistakes

- Not splitting at midpoint.
- Forgetting probability equals length ratio under uniformity.

Pattern Recognition

Min/max + symmetric interval ⇒ split at midpoint.

Question 2

Concept Being Tested

Bayes' theorem with continuous densities.
Equal posterior probabilities imply equal error.

Given

$$X|W \sim N(4, 4), \quad \sigma_1 = 2$$

$$X|B \sim N(6, 9), \quad \sigma_2 = 3$$

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observed:

$$X = 5$$

Equal error condition:

$$P(B|X=5) = P(W|X=5) = \frac{1}{2}$$

Bayes' Formula

$$P(B|X=5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)}$$

Set equal to $1/2$.

Normal Density Used

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solve

$$6\alpha = 3\alpha + 2(1-\alpha)e^{-5/72}$$

$$3\alpha = 2(1-\alpha)(0.93)$$

$$4.86\alpha = 1.86$$

$$\boxed{\alpha \approx 0.3827}$$

Common Mistakes

- Ignoring priors.
- Cancelling exponentials improperly.

Recognition Pattern

Equal posterior ⇒ numerator equals denominator.

Question 3

(a) Uniform Case

$$X \sim Uniform(0, A)$$

$$f_X(x) = \frac{1}{A}$$

Minimize:

$$E[|X - a|]$$

Split:

$$E[|X - a|] = \int_0^a (a - x)\frac{1}{A}dx + \int_a^A (x - a)\frac{1}{A}dx$$

After evaluation:

$$\frac{2a^2 - 2aA + A^2}{2A}$$

Differentiate:

$$\frac{1}{2A}(4a - 2A) = 0$$

$$4a = 2A$$

$$\boxed{a = \frac{A}{2}}$$

(b) Exponential Case

$$X \sim Exponential(\lambda)$$

$$f(x) = \lambda e^{-\lambda x}$$

Derivative:

$$\frac{d}{da}E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = \frac{1}{2}$$

$$\boxed{a = \frac{\ln 2}{\lambda}}$$

Insight

Uniform ⇒ midpoint.
Exponential ⇒ solve derivative equation.

Question 4

Concept

Mixture of exponentials via law of total probability.

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$\boxed{P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}}$$

Question 5

Concept

Conditional Poisson distribution.

$$P(X = k | X + Y = n) = \frac{P(X = k) P(Y = n - k)}{P(X + Y = n)}$$

After simplification:

$$\boxed{\binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}}$$

Recognition

Conditional Poisson ⇒ Binomial structure.

Question 6

Concept

Transformation of Uniform variable with mixed distribution.

$$X \sim Uniform[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

CDF:

$$F_Y(y) = \begin{cases} 0 & y < -2 \\ \frac{y+2}{4} & -2 \leq y < -1 \\ \frac{1}{4} & -1 \leq y < 0 \\ \frac{3}{4} & 0 < y < 1 \\ \frac{y+2}{4} & 1 \leq y \leq 2 \\ 1 & y > 2 \end{cases}$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

PDF:

$$f_Y(y) = \frac{1}{4} \text{ on } (-2, -1), (1, 2)$$

Question 7

$$S_{1000} = S_0 u^k d^{1000-k}$$

Condition:

$$u^k d^{1000-k} \geq 1.30$$

Taking logs:

$$k(\ln u - \ln d) + 1000 \ln d \geq \ln(1.30)$$

$$k \geq 470$$

$$k \sim Binomial(1000, 0.52)$$

$$P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Question 8

$$X \sim Exponential(1/2)$$

$$f(x) = \frac{1}{2}e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geq 10 | X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Question 9

$$Z = \sqrt{X^2 + Y^2}$$

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Using:

$$\frac{d}{dz}\sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final PDF:

$$f_Z(z) = z \int_{-z}^{z} \frac{f_X(x)\left[f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})\right]}{\sqrt{z^2 - x^2}} dx$$

Question 10

$$M_n = \max(X_1, \ldots, X_n)$$

$$P(M_n \leq x) = F(x)^n$$

Joint CDF:

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y \\ F(y)^{n+1}, & x > y \end{cases}$$

PHASE 2 — APPENDIX

Core Distributions Used

Uniform:

$$f(x) = \frac{1}{A}$$

Exponential:

$$f(x) = \lambda e^{-\lambda x}$$

Normal:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Poisson:

$$e^{-\lambda} \frac{\lambda^k}{k!}$$

Binomial:

$$\binom{n}{k} p^k (1-p)^{n-k}$$

Structural Results Used

- Probability via interval length (Uniform).
- Bayes posterior equality.
- Minimization via derivative.
- Law of total probability for mixtures.
- Conditional Poisson ⇒ Binomial.
- Mixed discrete-continuous transformation.
- Log transformation of multiplicative inequalities.
- Maximum of i.i.d. ⇒ power of CDF.

This completes the fully structured, strictly material-based lecture scribe from Tutorial-2 Tutorial_2_Solution.

If you want, I can now compress this into an ultra-compact 2–3 page "Night Before Exam" sheet using only these results.

Sources